Design Quiz
09 Connect 4

Technical Requirements

List the technical requirements you have identified in the instructions. Remember, these are requirements that need to be met to earn full autograder credit, but that don't affect the program output.

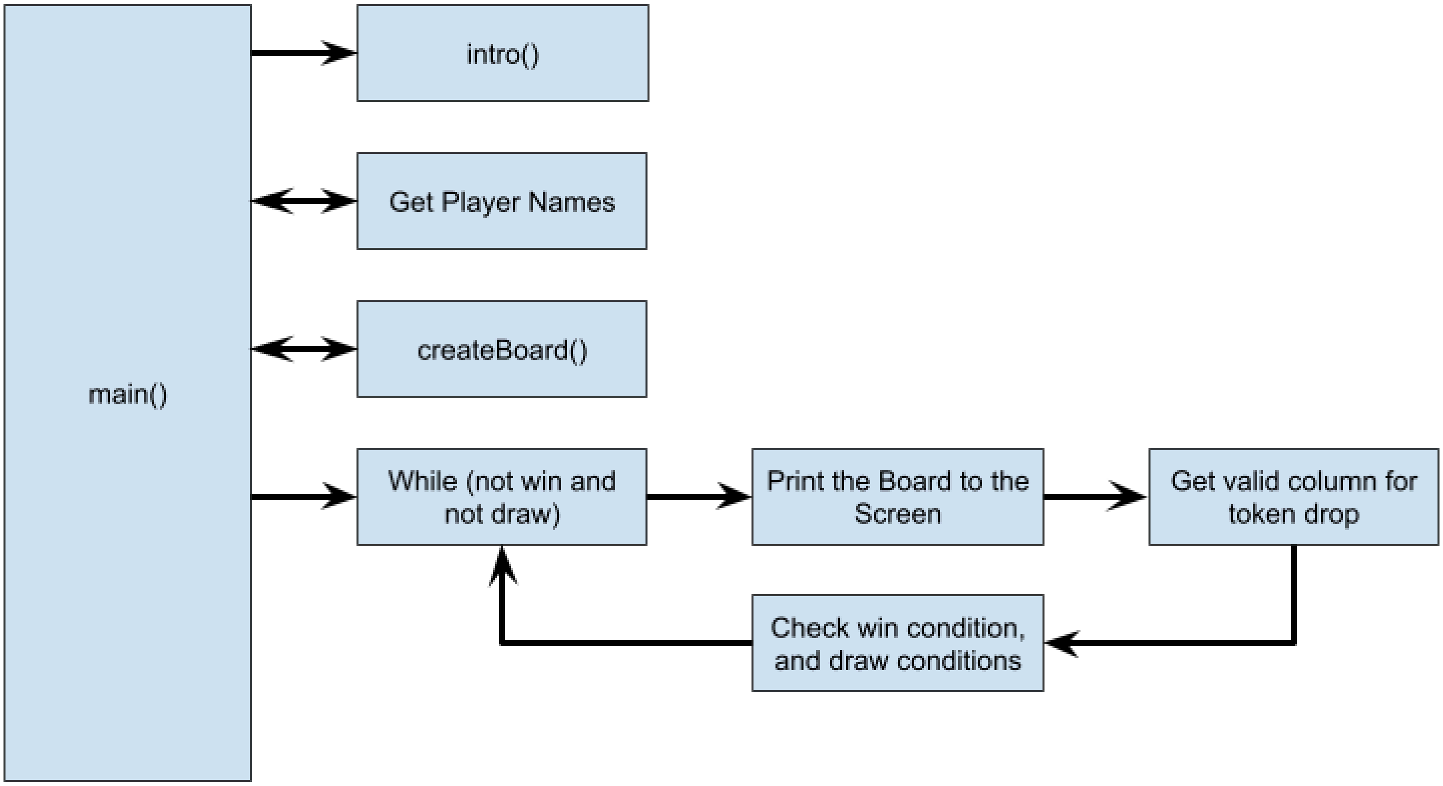
- Use a 2D array to store the game state
- You can't use a class variable
- You must use helper methods provided
- You can't use any array greater than 2D.
- Check if the one player wins or both draws.
- You must read in the input using the nextLine() method.
- Make sure that column inputs are in between 1-7

High-Level Summary

Write a high-level summary of the solution in English words. This should correspond, more or less, to the main() method.

- Read both player names.
- Create an empty board.
- Loop through each turns:
    - Ask the current player for the column that want to place the token into it.
    - Checks to see if the column played is valid.
    - Print the board to the console again.
    - Check to see if either player won or both tied.
- Print the Results at the end of the game.

Program Flow



Variables and Data Structures

Specify how key data will be stored in your program.

Variable Type and/or Data Structure	Data
char[][] board - Row index: A row value from 0-5 - Column Index: A Column Value from 0-6 - Value: Tells the program whether the value at that position is empty, taken by the red user, or taken by the blue user.	the board with rows and columns and what is at each row/col intersection (nothing, player 1 checker, or player 2 checker)

Key Algorithms

Consider an initial approach to determine if the game is over or not. The game is over when one player has 4 checkers in a row or when all board spaces have checkers.

- Loop through all the elements in the array.
    - Check the token at the position to see if the token is not '.'
        - Check if it is possible to win going right
            - Check if the win condition is true.

- Check if is possible to win going down
    - Check if the win condition is true
- Check if it possible to win going diagonally
    - Check if the win condition is true
- Check Draw Condition